E3b: E3 at larger B, and F4 run to full solves

Extension of `E3.md`, run under `E3b-PROTOCOL.md` (commit `f9cf6ab116`, frozen before any measured run). No security claim. The curves are a restricted prime family (`2^t | p − 1`). Nothing here concerns P-192 or P-224.

What E3 left open, and the answer

E3 (PR #2323) compared the Semaev/F4 oracle on the `prime-kummer` base with `known-log:scalars=random` + MITM at equal B, but stopped at B = 19 and priced its F4 arms from the grid arm's records. E3b closes both limits as far as this host allows.

1. **Larger B (Part L).** The two arms that run end to end, the `V³` grid reference on the Kummer base and known-log MITM (`m = 3`), at tower heights `t = 4…9` (B = 6 to 276) on three new 24-bit prime-order curves, 8 targets each, 312 runs, all verified. This passes the known-log arm's own optimum.
   - **Known-log MITM is cheapest at `t = 7` (B = 61–69)** on every curve, at 3.2–4.4× `rho-neg` (w = 0).
   - **The order of the grid and known-log MITM flips inside the band at every t**, and no crossover in B exists. At w = 0 the grid's point estimate is below known-log at every t (ratio 0.36–0.75 at `t = 4`, B = 6–10, falling to 0.056–0.061 at `t = 9`, B ≈ 260); the 90% interval excludes 1 in 15 of the 18 cells. At w = 0.1 and w = 1 it is 239× to $7.3 \cdot 10^5$× dearer, and the gap widens with B.
   - **The break-even weight w^* stops falling.** It falls to about 10^{-6} by `t = 7` and then stays at $1.2–1.7 \cdot 10^{-6}$ for `t = 7, 8, 9` (narrow intervals). E3′s prediction that it keeps falling was for F4 and is not tested at `t ≥ 6`, because F4 was not run there.
   - **Against rho the order also flips inside the band.** At w = 0.1 and 1 the grid is $1.9 \cdot 10^4$ to $1.6 \cdot 10^6$× `rho-neg` (w = 0.1) and $1.1 \cdot 10^5$ to $9.6 \cdot 10^6$× (w = 1). At w = 0 it is 0.78–1.70× at `t ≥ 7` and *below* rho at `t = 8` on e3-c5 and e3-c6 (the 90% interval excludes 1 only on e3-c6), where w = 0 prices all of the grid's solving work at nothing. Under AGENTS.md §5a that is unresolved, and it is not an improvement.
2. **F4 run to full solves (Part F): closed at `t = 2` only.** At `t = 2` (both presentations, targets 0 and 1) the measured cost equals E3′s substitution to five decimals at every weight. This is the weak case for the check: only 3 of 434 258 calls were consistent systems. **`t = 3` was not completed.** Its runs died twice with the host session (deviations D1, D2), and the work they need is about 46–220 h for the chain arm on this host, above the spec's 72 h timeout (see "Why `t = 3` was not completed"). So E3′s substitution is now checked against full F4 solves on a 24-bit curve at `t = 2` and on a 12-bit control, and not at any `t ≥ 3`.

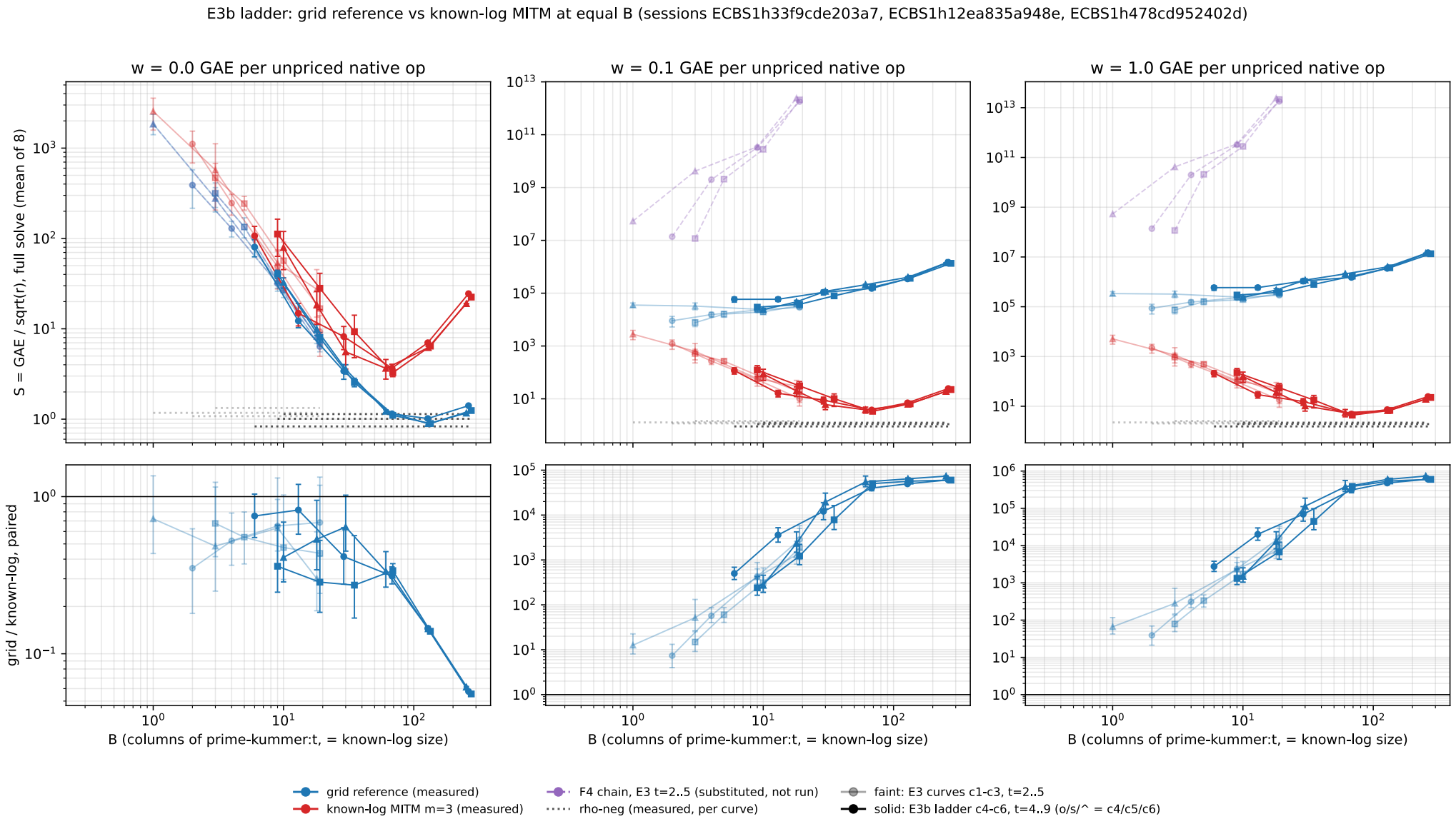


Figure 1: E3b ladder

Figure 1. Top: S = GAE/√r of a full solve against B at w = 0, 0.1 and 1; solid = E3b (curves e3-c4/c5/c6, markers o/s/^), faint = E3 (c1/c2/c3, `t = 2…5`); blue = grid reference, red = known-log MITM, dashed purple = F4 chain, ***substituted, not run*** *(from E3), dotted grey = `rho-neg` per curve. Bottom: grid / known-log paired by target. 8 targets per point, 90% bootstrap intervals. Sessions ECBS1h33f9cde203a7, ECBS1h12ea835a948e, ECBS1h478cd952402d.*

What ran

item	value
curves	e3-c4 `p = 10398721`, e3-c5 `p = 15052801`, e3-c6 `p = 15161857`: 24-bit, `a = −3`, prime order, 2^9 divides p − 1 (table in the protocol)
ladder arms	per curve and `t = 4…9`: `sem-grid-t{t}` (`prime-kummer:t={t}` + `pkm-semaev:m=3,engine=exhaustive`), `kl-mitm3-t{t}` (`known-log:size=B,scalars=random` + `mitm:m=3,negation_folded=1`), plus `rho-neg`; 8 `planted_shared` targets per curve, target seed 20261019, 1 round, no warm-up
ladder sessions	e3-c4 `ECBS1h33f9cde203a7`, e3-c5 `ECBS1h12ea835a948e`, e3-c6 `ECBS1h478cd952402d`: 312 of 312 runs, all `verified`
F4 full-solve arms	planned: `sem-chain-t2`, `sem-full-t2` (targets 0, 1), `sem-chain-t3` (target 0), `sem-full-t3` (targets 0, 1), E3′s oracle strings with `crosscheck=1` and no `max_trials`, on E3′s e3-c1 and E3′s targets. **Completed: the `t = 2` arms only**
F4 sessions	`t = 2`: chain `ECBS1h305ebf1e98d2`, full `ECBS1hbe41d11e8904` (4 of 4 runs each, all `verified`). `t = 3`: none completed (D1, D2)
binary	the E3 build, `ecbench` sha256 `b72458fbfa802e88…`, the binary of every E3 record
host	macOS arm64, shared, heavily loaded, no CPU pinning (`--cpus none`), isolation L0. Operation counts only; wall time is a practicality note
aggregation	`examples/generator_effect_e3_summary.rs` (extended additively: E3′s `summary.json` is reproduced exactly) ⟶ `e3b/summary.json`; `examples/generator_effect_e3b_f4check.rs` ⟶ `e3b/f4check.json`; 10 000 resamples, seed 20261019 (E3b's added statistics, seed 20261020); pricing through `ecbench::band`
records	`e3b/sessions/`, specs in `e3b/specs/`, sizes in `e3b/sizes.jsonl`
graphs	`e3b/e3b_ladder.svg`, `e3b/e3b_f4.svg` (PNG beside them), from `e3b/plot_e3b.py`, which plots the two JSON files only

`S(w) = (G − C + w·(U + n_C)) / √r` as in E3, with `C = n_C = 0` (fresh curves, no pinned calibration).

Results, Part L

1. Yield at equal B (L1 fails by one cell)

N_3 of the two bases agree within 0.3% in every cell (ratio known-log / Kummer 0.9985–1.0030). The Wilson 90% interval of the measured hit rate covers $h = N_3/r$ in **33 of 36** (curve, t, arm) cells. The three misses are e3-c4 `t = 5` for both arms (grid $3.61\cdot10^{-4}$ [3.08, 4.23]$\cdot10^{-4}$ and known-log $5.8\cdot10^{-4}$ [3.3, 10.4]$\cdot10^{-4}$ against $h = 2.85\cdot10^{-4}$) and e3-c4 `t = 9` for the grid (0.892 [0.880, 0.902] against 0.903). L1 required at least 34, so it **fails**. 90% intervals are expected to miss about 3.6 of 36 cells by chance, which is what happened, so this is read as a sampling effect and not a defect in the oracle. The known-log arm stops at its first relation, so its trials are geometric waiting times and a fixed-n interval is only approximate for them.

Hit rates and exact counts (Wilson 90% intervals):

curve	t	B	N3 Kummer	N3 known-log	h = N3/r	grid hit rate [Wilson 90%]	known-log hit rate [Wilson 90%]	grid trials	kl trials
e3-c4	4	6	304	304	2.922e-05	2.480e-05 [1.97e-05, 3.12e-05]	3.793e-05 [2.14e-05, 6.73e-05]	2056689	210898
e3-c4	5	13	2964	2964	2.849e-04	3.607e-04 [3.08e-04, 4.23e-04]	5.832e-04 [3.29e-04, 1.03e-03]	296685	13717
e3-c4	6	29	32518	32534	3.126e-03	2.814e-03 [2.51e-03, 3.15e-03]	2.500e-03 [1.41e-03, 4.43e-03]	73558	3200
e3-c4	7	68	411836	411584	3.959e-02	3.822e-02 [3.56e-02, 4.10e-02]	3.463e-02 [1.96e-02, 6.04e-02]	13422	231
e3-c4	8	128	2457910	2460400	2.363e-01	2.422e-01 [2.31e-01, 2.54e-01]	2.353e-01 [1.38e-01, 3.72e-01]	3943	34
e3-c4	9	263	9390715	9396807	9.027e-01	8.917e-01 [8.80e-01, 9.02e-01]	1.000e+00 [7.47e-01, 1.00e+00]	2114	8
e3-c5	4	9	996	996	6.614e-05	6.321e-05 [5.25e-05, 7.61e-05]	4.391e-05 [2.47e-05, 7.79e-05]	1233909	182201
e3-c5	5	19	9196	9196	6.107e-04	6.322e-04 [5.52e-04, 7.25e-04]	3.649e-04 [2.06e-04, 6.47e-04]	229354	21925
e3-c5	6	35	57166	57216	3.796e-03	4.110e-03 [3.72e-03, 4.54e-03]	2.176e-03 [1.23e-03, 3.86e-03]	64967	3676
e3-c5	7	69	430302	431594	2.857e-02	2.809e-02 [2.61e-02, 3.02e-02]	3.077e-02 [1.74e-02, 5.38e-02]	18086	260
e3-c5	8	134	2889312	2892498	1.919e-01	1.990e-01 [1.90e-01, 2.08e-01]	1.739e-01 [1.01e-01, 2.83e-01]	4996	46
e3-c5	9	276	12716067	12719539	8.444e-01	8.518e-01 [8.39e-01, 8.64e-01]	7.273e-01 [4.80e-01, 8.85e-01]	2314	11
e3-c6	4	10	1360	1360	8.969e-05	8.348e-05 [6.97e-05, 1.00e-04]	6.838e-05 [3.85e-05, 1.21e-04]	994242	116993
e3-c6	5	18	7824	7824	5.160e-04	4.788e-04 [4.16e-04, 5.51e-04]	5.346e-04 [3.01e-04, 9.49e-04]	286160	14964
e3-c6	6	30	35990	36002	2.373e-03	2.356e-03 [2.11e-03, 2.63e-03]	3.279e-03 [1.85e-03, 5.81e-03]	96794	2440
e3-c6	7	61	298996	299670	1.972e-02	2.016e-02 [1.87e-02, 2.18e-02]	1.677e-02 [9.48e-03, 2.95e-02]	22417	477
e3-c6	8	129	2611328	2607494	1.722e-01	1.785e-01 [1.70e-01, 1.87e-01]	1.212e-01 [6.97e-02, 2.03e-01]	5463	66
e3-c6	9	253	11521250	11514803	7.598e-01	7.580e-01 [7.43e-01, 7.72e-01]	8.000e-01 [5.41e-01, 9.31e-01]	2413	10

2. Known-log MITM's optimum (L2 holds)

Mean S at w = 0, B at its minimum: e3-c4 3.67 (B = 68), e3-c5 3.26 (B = 69), e3-c6 3.65 (B = 61), all at `t = 7`, and above `rho-neg` at every t on every curve (3.2–4.4× at the minimum; intervals in the table). Its cost is higher on both sides of this t in every curve (the ladder has one point per t).

3. S at the band weights (L3 holds)

Full table at w = 0, with the break-even weight (S in GAE/√r, mean of 8; intervals are 90% bootstrap, ratios paired by target):

curve	t	B	S grid	S known-log	S rho	grid / kl [90%]	grid / rho [90%]	kl / rho	w*
e3-c4	4	6	80.41	106.79	0.83	0.753 [0.549, 1.037]	96.73 [65.56, 143.44]	128.47	4.5e-05
e3-c4	5	13	12.23	14.88	0.83	0.822 [0.576, 1.196]	14.71 [11.34, 20.45]	17.90	4.5e-06
e3-c4	6	29	3.41	8.20	0.83	0.415 [0.277, 0.634]	4.10 [3.54, 4.85]	9.87	4.4e-06
e3-c4	7	68	1.15	3.67	0.83	0.313 [0.278, 0.357]	1.38 [1.16, 1.74]	4.41	1.7e-06
e3-c4	8	128	1.02	7.01	0.83	0.145 [0.139, 0.150]	1.22 [1.01, 1.56]	8.44	1.7e-06
e3-c4	9	263	1.41	24.48	0.83	0.058 [0.057, 0.058]	1.70 [1.40, 2.19]	29.45	1.6e-06
e3-c5	4	9	40.35	111.94	1.01	0.361 [0.247, 0.658]	39.97 [31.12, 54.54]	110.88	2.4e-04
e3-c5	5	19	8.02	28.09	1.01	0.286 [0.184, 0.526]	7.95 [6.93, 9.55]	27.83	5.3e-05
e3-c5	6	35	2.55	9.31	1.01	0.273 [0.169, 0.565]	2.52 [1.97, 3.40]	9.22	8.5e-06
e3-c5	7	69	1.10	3.26	1.01	0.338 [0.304, 0.374]	1.09 [0.85, 1.46]	3.23	1.3e-06
e3-c5	8	134	0.89	6.43	1.01	0.139 [0.135, 0.143]	0.89 [0.71, 1.17]	6.37	1.5e-06
e3-c5	9	276	1.25	22.42	1.01	0.056 [0.055, 0.056]	1.23 [0.98, 1.62]	22.21	1.6e-06
e3-c6	4	10	32.52	79.31	1.14	0.410 [0.287, 0.688]	28.55 [22.52, 34.98]	69.63	2.0e-04
e3-c6	5	18	9.80	18.31	1.14	0.535 [0.342, 0.947]	8.61 [7.14, 10.48]	16.07	1.8e-05
e3-c6	6	30	3.56	5.56	1.14	0.641 [0.450, 1.021]	3.13 [2.77, 3.65]	4.88	1.7e-06
e3-c6	7	61	1.22	3.65	1.14	0.335 [0.265, 0.447]	1.07 [0.90, 1.26]	3.21	1.2e-06
e3-c6	8	129	0.89	6.19	1.14	0.144 [0.141, 0.148]	0.78 [0.66, 0.92]	5.43	1.3e-06
e3-c6	9	253	1.17	19.00	1.14	0.061 [0.061, 0.062]	1.02 [0.87, 1.21]	16.68	1.3e-06

- **w = 0.** The grid's point estimate is below known-log at all 18 cells; the interval excludes 1 in 15 and contains it in 3 (e3-c4 `t = 4, 5`; e3-c6 `t = 6`). At w = 0 the grid pays only target draws and lifts, about one group addition a trial, while known-log MITM pays for a pair table of about B^2 points.
- **w = 0.1 and w = 1.** The grid is above known-log in every cell (row intervals in the table below), by at least 239× (w = 0.1) and $1.3 \cdot 10^3$ (w = 1) at `t = 4`, at least $7.8 \cdot 10^3$ at `t = 6`, rising to $6.0–7.4 \cdot 10^4$ (w = 0.1) and $6.0–7.3 \cdot 10^5$ (w = 1) at `t = 9`:

curve	t	grid / kl at w = 0.1	grid / kl at w = 1	grid / rho at w = 0.1	kl / rho at w = 0.1
e3-c4	4	499 [366, 687]	2.75e+03 [2.02e+03, 3.79e+03]	6.49e+04	130
e3-c4	5	3.61e+03 [2.49e+03, 5.23e+03]	2.01e+04 [1.39e+04, 2.98e+04]	6.51e+04	18.1
e3-c4	6	1.21e+04 [7.89e+03, 1.88e+04]	6.96e+04 [4.53e+04, 1.11e+05]	1.2e+05	9.89
e3-c4	7	4e+04 [3.48e+04, 4.71e+04]	3.11e+05 [2.61e+05, 3.87e+05]	1.68e+05	4.2
e3-c4	8	4.96e+04 [4.63e+04, 5.34e+04]	4.79e+05 [4.4e+05, 5.21e+05]	3.86e+05	7.8
e3-c4	9	6.05e+04 [5.87e+04, 6.22e+04]	6.05e+05 [5.87e+05, 6.21e+05]	1.64e+06	27.1
e3-c5	4	239 [164, 430]	1.32e+03 [898, 2.36e+03]	2.68e+04	112
e3-c5	5	1.23e+03 [783, 2.23e+03]	6.8e+03 [4.3e+03, 1.24e+04]	3.44e+04	28
e3-c5	6	7.81e+03 [4.75e+03, 1.63e+04]	4.48e+04 [2.67e+04, 9.81e+04]	7.2e+04	9.22
e3-c5	7	5.02e+04 [4.4e+04, 5.68e+04]	3.86e+05 [3.29e+05, 4.52e+05]	1.54e+05	3.07
e3-c5	8	5.68e+04 [5.32e+04, 6.04e+04]	5.42e+05 [4.97e+05, 5.87e+05]	3.34e+05	5.88
e3-c5	9	6.01e+04 [5.94e+04, 6.07e+04]	5.99e+05 [5.92e+05, 6.05e+05]	1.22e+06	20.4
e3-c6	4	272 [189, 456]	1.5e+03 [1.04e+03, 2.53e+03]	1.9e+04	70.1
e3-c6	5	2.34e+03 [1.49e+03, 4.22e+03]	1.3e+04 [8.23e+03, 2.35e+04]	3.78e+04	16.2
e3-c6	6	1.95e+04 [1.37e+04, 3.08e+04]	1.14e+05 [7.81e+04, 1.88e+05]	9.47e+04	4.86
e3-c6	7	5.46e+04 [4.27e+04, 7.4e+04]	3.8e+05 [2.82e+05, 5.59e+05]	1.69e+05	3.09
e3-c6	8	6.42e+04 [6.1e+04, 6.75e+04]	5.99e+05 [5.66e+05, 6.33e+05]	3.22e+05	5.02
e3-c6	9	7.37e+04 [7.24e+04, 7.49e+04]	7.35e+05 [7.22e+05, 7.47e+05]	1.13e+06	15.3

- **§5a verdict.** The order of the grid and known-log MITM changes between w = 0 and w = 0.1 at every (curve, t), so the order does not hold across the band. **Unresolved. Not a win for either.**

4. Crossover in B (L4 holds, with a correction to E3′s trend)

- **No `B*` exists at any weight.** The ratio grid / known-log is below 1 at every t for w = 0 and above 1 at every t for w = 0.1 and 1.
- **`w*` is below 10^{-3} at every `t ≥ 6`** (largest $8.5 \cdot 10^{-6}$, e3-c5 `t = 6`).
- **`w*` does not keep falling.** From `t = 7` it is flat at $1.2–1.7 \cdot 10^{-6}$ on all three curves (90% intervals up to about $\pm 0.4 \cdot 10^{-6}$ at `t = 7`, $\pm 0.1 \cdot 10^{-6}$ or narrower at `t = 8, 9`). The grid's multiplications per call grow 6.9–7.9× per level (7 440 at `t = 4`, 180 797 952 at `t = 9`) while the exact `h` grows 3.8–13× per level (slower past `t = 7`, where it saturates toward 1). Why `w*` levels off is not derived here; the plateau is the measured fact.
- **What this does and does not say about F4.** E3′s F4 `w*` fell 49–880× per level over `t = 2…5`. F4 was not run at `t ≥ 6` (one `t = 6` system took 44 minutes and 14 GB in the PKM note §16). Extending E3′s measured growth of F4′s cost (67–406× per level) puts F4 far above the grid at every `t ≥ 6`, and so above known-log at every w > 0 there. **That is an extrapolation, not a measurement.** No F4 number at `t ≥ 6` appears in any table or figure of this report.

5. Against rho (L5 fails)
The prediction was that the grid stays above `rho-neg` at w = 0 at every t, and is closest at `t = 9`. Measured, grid / rho-neg at w = 0, mean of 8 with 90% interval:

curve	t = 7	t = 8	t = 9
e3-c4	1.38 [1.16, 1.74]	1.22 [1.01, 1.56]	1.70 [1.40, 2.19]
e3-c5	1.09 [0.85, 1.46]	**0.89 [0.71, 1.17]**	1.23 [0.98, 1.62]
e3-c6	1.07 [0.90, 1.26]	**0.78 [0.66, 0.92]**	1.02 [0.87, 1.21]

- At `t = 8` the grid is below rho on e3-c5 and e3-c6; the interval excludes 1 only on e3-c6. The closest approach to rho is at `t = 8` on every curve, not at `t = 9`. **L5 fails on both counts.**
- The part of L5 that holds: at w = 0.1 the grid is $1.9{\cdot}10^4$ to $1.6{\cdot}10^6$ times rho and at w = 1 it is $1.1{\cdot}10^5$ to $9.6{\cdot}10^6$ times (smallest at `t = 4`–`5`, largest at `t = 9`).
- **What w = 0 means.** It prices the grid's $2.3{\cdot}10^7$ field multiplications per call at `t = 8` ($1.8{\cdot}10^8$ at `t = 9`) at zero. On e3-c6 at `t = 8` the grid's S rises by about $4.0{\cdot}10^6$ per unit w (derived from its S at w = 0.1), so a price above about $6{\cdot}10^{-8}$ GAE per multiplication puts it back above rho, and at w = 0.1 it is $3.2{\cdot}10^5\times$ rho. Rho is ahead across the rest of the band. Against rho too the order flips inside the band, so it is unresolved under §5a. This is a statement about the pricing rule, not a faster algorithm.
- Known-log MITM is above rho at every t and w on every curve (interval excludes 1 in all 54 cells): 3.1–130× at w = 0.1 and 2.3–138× at w = 1.

Results, Part F

F4 against E3′s substitution at `t = 2`
Four full F4 solves on e3-c1 (chain and full presentation, E3′s targets 0 and 1), each run to a verified recovery:

arm	target	trials	relations	F4 calls	F4 mult-adds per call	E3 capped mean	measured / substituted U
`sem-chain-t2`	0	444 341	3	434 258	$3.9079{\cdot}10^5$	$3.9079{\cdot}10^5$	1.00000
`sem-chain-t2`	1	841 257	3	805 655	$3.9079{\cdot}10^5$	$3.9079{\cdot}10^5$	1.00000
`sem-full-t2`	0	444 341	3	434 258	$2.2429{\cdot}10^5$	$2.2429{\cdot}10^5$	1.00000
`sem-full-t2`	1	841 257	3	805 655	$2.2429{\cdot}10^5$	$2.2429{\cdot}10^5$	1.00000

- **F1 holds** for these runs: trials, relations and group GAE equal those of E3′s `sem-grid-t2` record of the same target exactly, `same_stream` is true in every row, and `pkm_crosscheck_disagreements` is 0 with 0 budget overruns.
- **F2 holds** for these runs: measured and substituted S agree to five decimals at w = 0, 0.1 and 1.
- The agreement is nearly forced at `t = 2`: only 3 of the 434 258 calls were consistent systems, so the refutation mean carries the sum, and a consistent system costs 0.5–1.5× a refutation. The check is exact on the three things it tests (same stream, per-call count by shape, accounting), and weak on the consistent-system cost at this t.
- Solver wall time (host shared and heavily loaded): chain 3.7 h and 1.5 h, full 3.4 h and 1.3 h for targets 0 and 1.

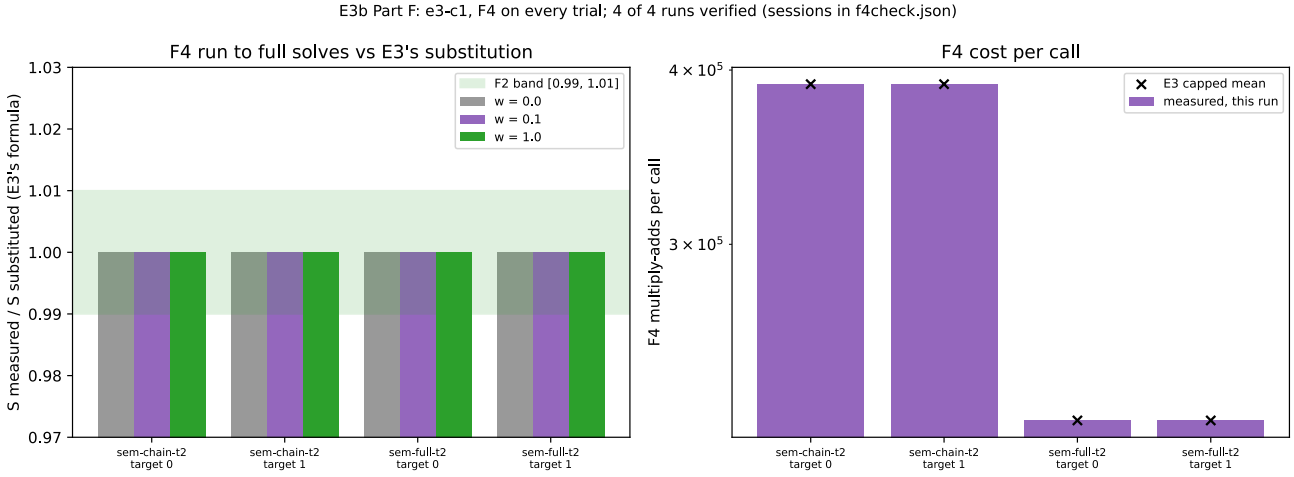


Figure 2: F4 against E3′s substitution at t = 2

Figure 2. e3-c1, `t = 2`, F4 run on every trial. Left: S measured / S substituted (E3′s formula) at w = 0, 0.1, 1 for the four runs; shaded band is F2′s [0.99, 1.01]. Right: F4 multiply-adds per call, measured (bars) against E3′s capped mean (crosses). Sessions ECBS1h305ebf1e98d2, ECBS1hbe41d11e8904; all 4 runs verified.

F4 against E3′s substitution at `t = 3`: not completed
No `t = 3` F4 record exists in this report. The runs were started twice (the launch and one rerun) and killed with the host session each time (D1, D2); the first interrupted `sem-full-t3` target 0 is not used (D1).

Why `t = 3` was not completed

Derived from recorded values, not measured at `t = 3`:

- The `t = 2` solves took longer per F4 call than E3′s capped runs: chain 30.6 ms (target 0) and 6.5 ms (target 1) per call against E3′s 1.64 ms (655 ms over 400 calls), i.e. 4.0× to 18.7×. Full: 28.2 and 5.9 ms against 1.41 ms, 4.2× to 20×. The spread is the load on a shared host.
- E3′s capped `t = 3` runs cost 108 ms (chain) and 55 ms (full) per call. A `t = 3` solve of target 0 makes about 381 500 F4 calls (the interrupted `sem-full-t3` run recorded 381 501, the same stream as the grid). That is 11.5 h (chain) and 5.8 h (full) at E3′s speed, and with the same 4–20× load factor about 46–220 h (chain) and 24–117 h (full) per target. The spec timeout is 72 h.
- Two sessions did not live that long (D1, D2), and a third in-session attempt is very unlikely to finish.

What is still needed: a detached launch that outlives the session, the E3 binary (sha256 `b72458fb…`), a timeout of several days, and ideally an idle host. `t = 3` is a follow-up, not a result.

Predictions against outcome

	prediction	outcome
L1	hit rates cover N_3/r in ≥ 34 of 36 cells; N_3 within 1%	**fails by one cell**: 33 of 36 (3 misses, all on e3-c4); N_3 within 0.3%: holds
L2	known-log lowest at `t = 7` on every curve; 1.5–10× rho there	holds: B = 68, 69, 61; 4.4×, 3.2×, 3.2×
L3	w = 0 grid below known-log at every t; ≥ 10^3 above at `t ≥ 6`, w ≥ 0.1; unresolved	holds in point estimates (interval excludes 1 in 15 of 18); smallest `t ≥ 6` factor $7.8{\cdot}10^3$
L4	no B*; $w^* < 10^{-3}$ at `t ≥ 6`; "w* falls"	no B* and $w^* < 10^{-3}$: hold; **w^* is flat from `t = 7`**, not falling
L5	grid above rho at w = 0 at every t, closest at `t = 9`, within 3×; ≥ 10^3 above at w ≥ 0.1	**fails**: below rho at `t = 8` on e3-c5 and e3-c6, closest at `t = 8`; the w ≥ 0.1 part holds
L6	F4 at `t ≥ 6` only "extrapolated, not run"	holds
F1	same stream as E3′s grid record; verified; 0 disagreements	`t = 2` holds (4 of 4). `t = 3`: not tested, interrupted twice (D1, D2)
F2	measured / substituted within [0.99, 1.01]	`t = 2` holds (all within $3.3{\cdot}10^{-6}$ of 1). `t = 3`: not tested (D1, D2)

Scope and limits

- **Population:** a restricted prime family (`2^t | p − 1`), three curves per part, `m = 3`, 24-bit group orders, 8 targets per curve. B up to 276 reaches the known-log arm's optimum, but for `m = 3` only, and not E1′s `m = 2` optimum of about 220–250, where the pair table is the whole cost.
- **Pricing:** w = 0 prices the grid's field multiplications at zero. Every statement about the grid at w = 0 depends on that choice.
- **F4:** run to full solves only at `t = 2` (this report) and on E3′s 12-bit control, and as capped runs at `t ≤ 5` in E3. At `t ≥ 3` E3′s F4 full-solve cost remains a substitution. At `t ≥ 6` F4 is not run and the report makes no F4 claim there beyond the labelled extrapolation.
- **The grid is a reference, not a proposed algorithm.** It decides the same systems as F4 with fewer operations; it is not a security claim.
- **No security statement.** These are generic precomputation shapes and no exponent moves. This is not an improvement over the IC baseline: the grid is above known-log at every w > 0, and ahead of or behind rho depending on the price of unpriced work.

Interrupted runs (protocol deviations D1 and D2)

The host session ended twice while the `t = 3` F4 runs were in progress. The three ladder sessions and the `t = 2` F4 sessions had completed both times.

- **D1.** The first pair (`ECBS1hf02ca1ecbf63` chain, `ECBS1hdfef6661171e` full) were left `running` with partial records (full: target 0 of `sem-full-t3` verified, 389 357 trials, 381 501 F4 calls; chain: no F4 record).
- **D2.** The rerun pair (`ECBS1h345b571362e6` chain, `ECBS1hb680db5cb1a8` full) was started at about 18:08 PDT on 2026-10-25 and ended with the session before any F4 call finished. Each wrote only its `rho-neg` record (chain: target 0; full: target 0) and no F4 record.

All four partial directories are kept outside the repository (they are marked `running` with no sealed record hash and would not pass the audit of a sealed session) and no number of theirs enters any table. The D1 `sem-full-t3` target-0 record is not used either: choosing it now, after the rerun failed, would be a post-hoc choice. Part F at `t = 3` is therefore not reported.

Files

- `E3b-PROTOCOL.md`: the frozen protocol, with deviations D1 and D2.
- `reports/run-20261021T0512Z-e3b/`: the run report (observations only), with its PDF.
- `e3b/specs/`, `e3b/sessions/`, `e3b/sizes.jsonl`: specs, records, sizes.
- `e3b/summary.json`, `e3b/f4check.json`: aggregates.
- `e3b/plot_e3b.py`, `e3b/e3b_ladder.{svg,png}`, `e3b/e3b_f4.{svg,png}`.
- `examples/generator_effect_e3_summary.rs`: grid w* and crossover added.
- `examples/generator_effect_e3b_f4check.rs`: Part F aggregation.
- `E3b.pdf` (this report), built by `build_e3b_pdf.sh`.